# An Exact, Distribution-Free SURE for Shrinkage Toward $\nu I$

Jack Wang

October 26, 2026

## 1 Setup

Let $X_1, \dots, X_T \in \mathbb{R}^p$ be i.i.d. with mean zero (known), covariance $\Sigma$ and $\mathbb{E}\|X\|^4 < \infty$. No distributional form is assumed. Let

$$S = \frac{1}{T}\sum_{t=1}^{T} X_t X_t^\top, \qquad F = \nu I, \quad \nu = \frac{\operatorname{tr} S}{p}, \qquad \hat{\Sigma}_\alpha = \alpha F + (1-\alpha)S.$$

Because the mean is known, $\mathbb{E}S = \Sigma$, so Lemma 1 of Yi and Zou (2013) applies (see `sure_derivation.pdf`):

$$R(\alpha) = \mathbb{E}\|\hat{\Sigma}_\alpha - \Sigma\|_F^2 = \alpha^2 D + (1-2\alpha)V + 2\alpha C, \tag{1}$$

$$D = \mathbb{E}\|F - S\|_F^2, \qquad V = \sum_{i,j}\operatorname{Var}(s_{ij}), \qquad C = \sum_{i,j}\operatorname{Cov}(f_{ij}, s_{ij}).$$

Define the statistics

$$\hat{M} = \frac{1}{T}\sum_t \|X_t\|^4, \qquad \hat{U}_2 = \frac{1}{T(T-1)}\sum_{t\neq u}(X_t^\top X_u)^2, \qquad \hat{U}_3 = \frac{1}{T(T-1)}\sum_{t\neq u}\|X_t\|^2\|X_u\|^2.$$

They are cheap to compute because $\sum_{t,u}(X_t^\top X_u)^2 = T^2\operatorname{tr}(S^2)$ and $\sum_{t,u}\|X_t\|^2\|X_u\|^2 = T^2(\operatorname{tr} S)^2$, so

$$\hat{U}_2 = \frac{T\operatorname{tr}(S^2) - \hat{M}}{T-1}, \qquad \hat{U}_3 = \frac{T(\operatorname{tr} S)^2 - \hat{M}}{T-1}.$$

## 2 Results

**Proposition 1** (Unbiased estimates of $V$ and $C$)**.** *Under the setup above,*

$$\hat{V} = \frac{\hat{M} - \hat{U}_2}{T}, \qquad \hat{C} = \frac{\hat{M} - \hat{U}_3}{pT}$$

*satisfy* $\mathbb{E}\hat{V} = V$ *and* $\mathbb{E}\hat{C} = C$.

*Proof. The term $V$.* The entry $s_{ij} = \frac{1}{T}\sum_t x_{ti}x_{tj}$ is an average of i.i.d. terms, so $\operatorname{Var}(s_{ij}) = \frac{1}{T}\big(\mathbb{E}[x_i^2 x_j^2] - \sigma_{ij}^2\big)$. Summing over $i, j$, and using $\sum_{i,j} x_i^2 x_j^2 = \|X\|^4$ and $\sum_{i,j}\sigma_{ij}^2 = \operatorname{tr}(\Sigma^2)$,

$$V = \frac{\mathbb{E}\|X\|^4 - \operatorname{tr}(\Sigma^2)}{T}.$$

$\hat{M}$ is a sample mean of $\|X_t\|^4$, so $\mathbb{E}\hat{M} = \mathbb{E}\|X\|^4$. For $t \neq u$, independence gives $\mathbb{E}(X_t^\top X_u)^2 = \mathbb{E}\,\mathrm{tr}(X_t X_t^\top X_u X_u^\top) = \mathrm{tr}(\Sigma\Sigma)$, so $\mathbb{E}\hat{U}_2 = \mathrm{tr}(\Sigma^2)$.

*The term $C$.* Since $f_{ij} = \nu\,\delta_{ij}$,

$$C = \sum_i \mathrm{Cov}(\nu, s_{ii}) = \frac{1}{p}\,\mathrm{Cov}\Big(\mathrm{tr}\,S, \sum_i s_{ii}\Big) = \frac{1}{p}\,\mathrm{Var}(\mathrm{tr}\,S).$$

$\mathrm{tr}\,S = \frac{1}{T}\sum_t \|X_t\|^2$ is an i.i.d. average, so $\mathrm{Var}(\mathrm{tr}\,S) = \frac{1}{T}\big(\mathbb{E}\|X\|^4 - (\mathrm{tr}\,\Sigma)^2\big)$. For $t \neq u$, $\mathbb{E}\|X_t\|^2\|X_u\|^2 = (\mathrm{tr}\,\Sigma)^2$, so $\mathbb{E}\hat{U}_3 = (\mathrm{tr}\,\Sigma)^2$. □

**Corollary 1** (Exact SURE)**.** *For every fixed $\alpha$,*

$$\mathrm{SURE}(\alpha) = \alpha^2\|F - S\|_F^2 + (1 - 2\alpha)\hat{V} + 2\alpha\hat{C}$$

*satisfies $\mathbb{E}\,\mathrm{SURE}(\alpha) = R(\alpha)$. It is minimized at*

$$\hat{\alpha}_{\mathrm{SURE}} = \frac{\hat{V} - \hat{C}}{\|F - S\|_F^2},$$

*which we clip to $[0, 1]$.*

The observed $\|F - S\|_F^2$ is trivially unbiased for $D$, and SURE is linear in $(\|F - S\|^2, \hat{V}, \hat{C})$. The minimizer $\hat{\alpha}_{\mathrm{SURE}}$ is a ratio of estimates, so it is *not* unbiased for $\alpha^*$. Unbiasedness holds for the risk curve, not for its argmin.

**Proposition 2** (When shrinkage reduces risk)**.** *Let $\alpha^* = (V - C)/D$. Then $\alpha^* \geq 0$, and*

$$R(\alpha) < R(0) \iff 0 < \alpha < 2\alpha^*.$$

*Proof.* From (1), $R(\alpha) - R(0) = \alpha^2 D - 2\alpha(V - C) = \alpha D(\alpha - 2\alpha^*)$. Since $D > 0$, this is negative exactly when $\alpha$ lies strictly between 0 and $2\alpha^*$.

To see that $V \geq C$, apply Cauchy–Schwarz to $\mathrm{tr}\,S = \sum_i s_{ii}$: $\mathrm{Var}(\mathrm{tr}\,S) \leq \big(\sum_i \sqrt{\mathrm{Var}\,s_{ii}}\big)^2 \leq p\sum_i \mathrm{Var}(s_{ii})$. Hence $C = \mathrm{Var}(\mathrm{tr}\,S)/p \leq \sum_i \mathrm{Var}(s_{ii}) \leq V$. □

The sample analogue is that $\mathrm{SURE}(\alpha) < \mathrm{SURE}(0)$ if and only if $0 < \alpha < 2\hat{\alpha}_{\mathrm{SURE}}$. This is an exact algebraic statement about the estimated curve. The statement about the true risk uses $\alpha^*$, not $\hat{\alpha}$. This corrects the version in the progress log, which stated the condition with $\hat{\alpha}$.

**Proposition 3** (Ledoit–Wolf is SURE minus two finite-sample terms)**.** *The LW (2004) intensity toward $\nu I$ (eq. 13 in Chen et al., 2010) is*

$$\hat{\rho}_{\mathrm{LW}} = \frac{\sum_t \|X_t X_t^\top - S\|_F^2}{T^2\,\|F - S\|_F^2} = \frac{\frac{T-1}{T}\,\hat{V}}{\|F - S\|_F^2}, \qquad \textit{whereas} \qquad \hat{\alpha}_{\mathrm{SURE}} = \frac{\hat{V} - \hat{C}}{\|F - S\|_F^2}.$$

*Proof.* Expanding the square, $\sum_t \|X_t X_t^\top - S\|^2 = \sum_t \|X_t\|^4 - 2T\,\mathrm{tr}(S^2) + T\,\mathrm{tr}(S^2) = T(\hat{M} - \mathrm{tr}\,S^2)$. Substituting $\mathrm{tr}\,S^2 = \big(\hat{M} + (T-1)\hat{U}_2\big)/T$ gives $T \cdot \frac{T-1}{T}(\hat{M} - \hat{U}_2)$. Dividing by $T^2$ yields $\frac{T-1}{T}\hat{V}$. The denominators agree because $\mathrm{tr}(S^2) - (\mathrm{tr}\,S)^2/p = \|S - \nu I\|_F^2$. □

So LW and SURE share the denominator. They differ only in the numerator: LW multiplies $\hat{V}$ by $(T-1)/T$ and drops $\hat{C} = \widehat{\mathrm{Var}}(\operatorname{tr} S)/p$. Both differences vanish as $T \to \infty$, which is why LW is consistent. In small samples the $(T-1)/T$ factor dominates and LW under-shrinks relative to the oracle, as the simulation shows. (The script confirms the identity to 10 digits.)

*Remark* (Gaussian consistency check). Under Gaussianity, $\mathbb{E}\|X\|^4 = (\operatorname{tr}\Sigma)^2 + 2\operatorname{tr}(\Sigma^2)$. Then $V = \big(\operatorname{tr}\Sigma^2 + \operatorname{tr}^2\Sigma\big)/T$, $C = 2\operatorname{tr}\Sigma^2/(pT)$, and $D = \operatorname{tr}\Sigma^2 + V - \operatorname{tr}^2\Sigma/p - C$. Substituting,

$$\alpha^* = \frac{(1-2/p)\operatorname{tr}\Sigma^2 + \operatorname{tr}^2\Sigma}{(T+1-2/p)\operatorname{tr}\Sigma^2 + (1-T/p)\operatorname{tr}^2\Sigma},$$

which is exactly the Gaussian oracle in eq. (7) of Chen et al. (2010). The decomposition (1) therefore reproduces their oracle as a special case. Outside the Gaussian family, the kurtosis enters through $\mathbb{E}\|X\|^4$, and $\hat{M}$ estimates it directly.

## 3 Simulation

Code: `sims/sure_sim.py` (seed 20261026, about 20 s). $\Sigma$ is AR(1) with $r = 0.5$ and $p = 100$. The data are Gaussian or multivariate $t_{10}$ rescaled to covariance $\Sigma$. We use 10 degrees of freedom so that $\hat{V}$ and $\hat{C}$ have finite variance, which needs finite 8th moments. The true $V, C, D$ are in closed form, using $\mathbb{E}\|X\|^4 = \kappa\,(\operatorname{tr}^2\Sigma + 2\operatorname{tr}\Sigma^2)$ with $\kappa = 1$ (Gaussian) or $\kappa = (\nu-2)/(\nu-4)$ ($t_\nu$).

### Unbiasedness (Proposition 1)

20,000 replications per row. Every mean is within about 1.3 Monte Carlo standard errors of the truth.

| data | $p$ | $T$ | $V$ | mean $\hat{V}$ | s.e. | $C$ | mean $\hat{C}$ | s.e. |
|---|---|---|---|---|---|---|---|---|
| Gaussian | 10 | 5 | 23.156 | 23.223 | 0.094 | 0.6311 | 0.6330 | 0.0050 |
| Gaussian | 50 | 10 | 258.244 | 258.455 | 0.306 | 0.3298 | 0.3286 | 0.0013 |
| Gaussian | 100 | 40 | 254.144 | 254.173 | 0.105 | 0.0829 | 0.0828 | 0.0001 |
| $t_{10}$ | 10 | 5 | 31.926 | 32.142 | 0.338 | 1.5082 | 1.5114 | 0.0275 |
| $t_{10}$ | 50 | 10 | 347.074 | 345.813 | 1.754 | 2.1064 | 2.0800 | 0.0247 |
| $t_{10}$ | 100 | 40 | 340.241 | 339.157 | 0.861 | 0.9439 | 0.9386 | 0.0061 |

### SURE curve against the true risk (Goal 2)

Figure 1 has $p = 100$ and $T = 20$. The average of 4,000 SURE curves matches the closed-form $R(\alpha)$ to within 0.2% (Gaussian) and 0.1% ($t_{10}$) of $\max R$. A direct Monte Carlo of the loss lies on the same line. Individual curves (grey) are noisy, especially under heavy tails, but their minimizers cluster near $\alpha^*$.

### Comparison with LW, RBLW and OAS (Goal 4)

2,000 replications per $T$, with $p = 100$. "Oracle" uses the fixed $\alpha^*$ computed from the true $\Sigma$.

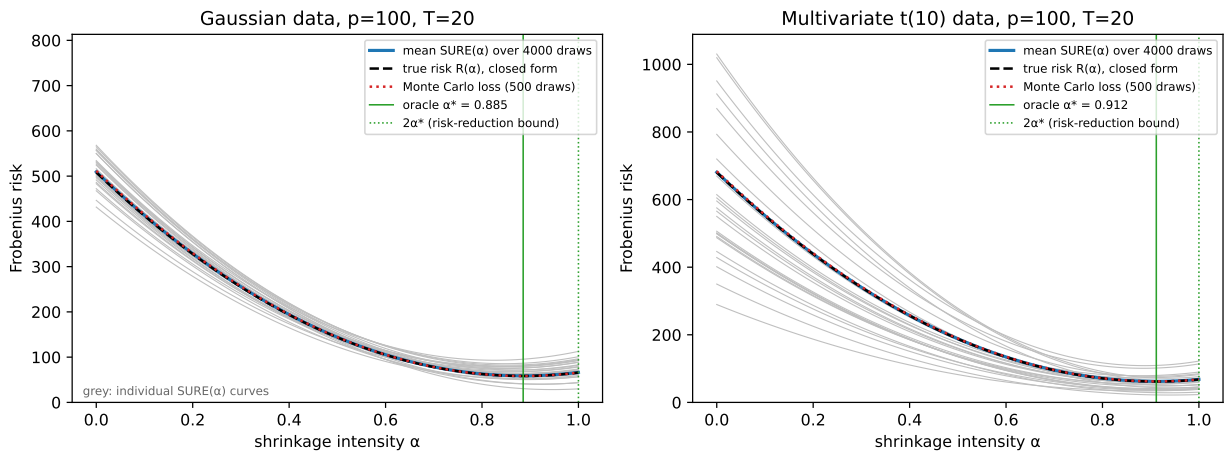



Figure 1: SURE($\alpha$) against the true Frobenius risk. Solid green: $\alpha^*$. Dotted green: $2\alpha^*$, the upper end of the risk-reduction interval (Prop. 2), here clipped at 1.

- **Gaussian data.** SURE and OAS are indistinguishable and stay within 3% of the oracle at every $T$. LW and RBLW under-shrink badly when $T$ is small ($\hat{\rho} \approx 0.80$ against $\alpha^* = 0.96$ at $T = 6$), giving 73% excess MSE. The cause is the factor $(T-1)/T$ in Proposition 3.
- $t_{10}$ **data.** SURE still tracks the oracle (at most 2.5% excess MSE; at $T \geq 50$ it matches or slightly beats the fixed-$\alpha^*$ oracle, which a data-dependent $\hat{\alpha}$ can do). RBLW and OAS plug in the Gaussian fourth moment, so they underestimate $V$ and shrink far too little. Their MSE is 1.2–4.6$\times$ the oracle and is still 18% above it at $T = 80$. LW is distribution-free, so it recovers as $T$ grows, but it is 1.87$\times$ the oracle at $T = 6$.

## 4 What this settles and what it does not

- Settled for the target $\nu I$ with known mean: an exact, distribution-free SURE exists (Prop. 1). The risk-reduction interval is $(0, 2\alpha^*)$ (Prop. 2). LW is SURE with two finite-sample terms removed (Prop. 3).
- Still open: the unknown mean. With $\bar{X}$ estimated, $\hat{M}, \hat{U}_2, \hat{U}_3$ need U-statistic corrections (Touloumis 2015 does this for $\operatorname{tr}\Sigma^2$ and $\operatorname{tr}^2\Sigma$).
- Still open: the targets $I$ and $D_S$ follow the same template ($C = 0$ for $I$; $C = \sum_i \operatorname{Var}(s_{ii})$ for $D_S$) but are not yet simulated.
- Still open: the single-index target. $F$ is nonlinear in $S$, so $C$ has no exact unbiased estimate. This is the main question for Rosenman.
- Robustness: only one $\Sigma$ (AR(1), $r = 0.5$) so far. Add $r \in \{0.1, 0.9\}$ and FBM to match Chen et al.'s experiments.

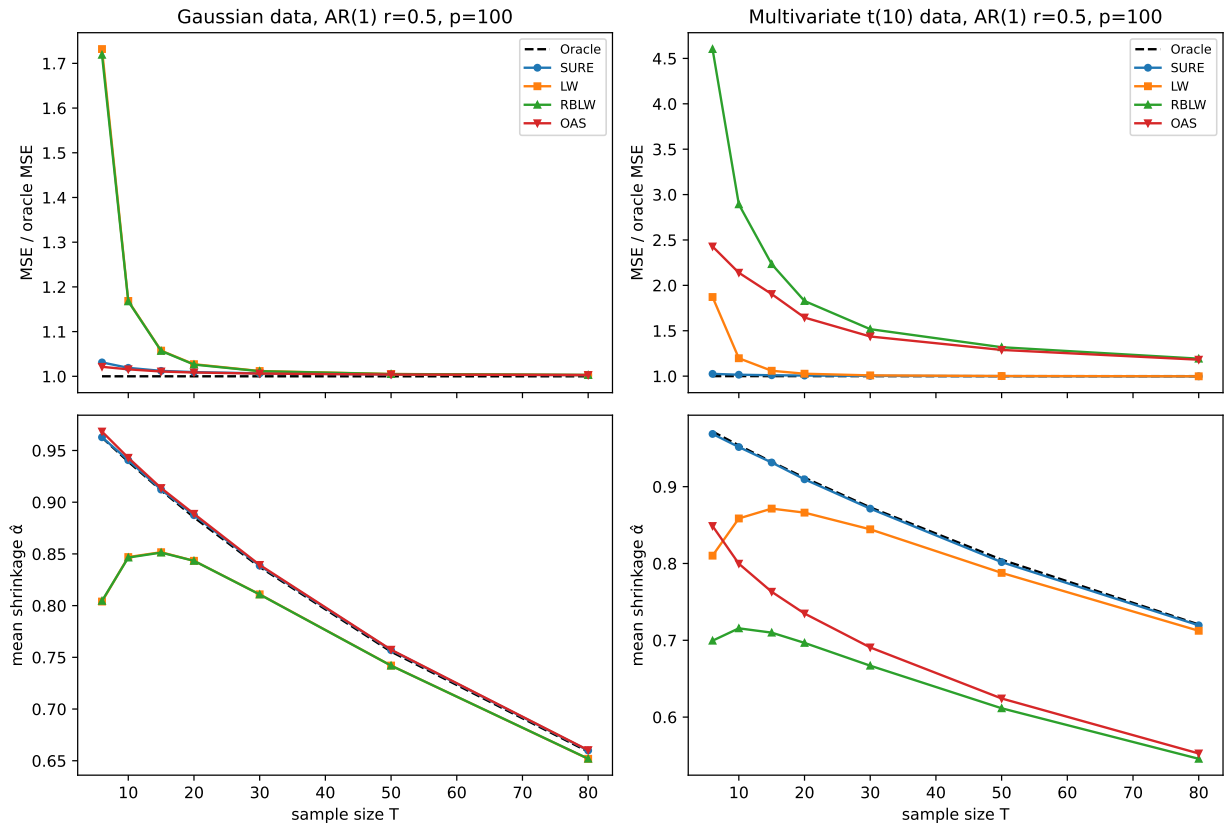



Figure 2: Top: Frobenius MSE relative to the oracle. Bottom: mean shrinkage intensity.